GROUP PROJECT: HOSPITAL READMISSION RISK - AN END-TO-END CLINICAL AI SOLUTION

Application of AI in Healthcare (MSAI-544) Residency - Group Project

PROJECT OVERVIEW

Duration: Saturday residency session, 8:00 AM to 7:30 PM with a 1.5-hour lunch break (10 hours of work time)

Type: Collaborative group project, one submission per group

Programming Language: Python AND/OR R (group choice; starter code provided for both)

Domain: Clinical risk prediction, responsible AI, and clinical integration

Points: 300

Project Goal

Your group has been engaged by a hospital system to build and evaluate an AI tool that identifies patients with diabetes who are at high risk of being readmitted within 30 days of discharge, and to advise the system on whether and how it should be deployed. You will take the problem from raw encounter data through cohort definition, exploratory analysis, feature engineering, modeling, explainability, a fairness audit, and a deployment plan, and deliver the result as a dashboard, an executive slide deck, and a technical report.

This project is the full life cycle of a clinical AI product compressed into one day. The modeling is the easy part; the decisions around it are what the hospital is paying for.

SCHEDULE (Saturday, 31 OCT 2026)

- **8:00-8:30 AM:** Project introduction and group formation
- **8:30 AM-12:00 PM:** Analyses 1-6 (cohort, exploration, features, baseline)
- **12:00-1:30 PM:** Lunch
- **1:30-4:30 PM:** Analyses 7-10 (models, explainability, fairness, deployment)
- **4:30-6:30 PM:** Dashboard, slide deck, technical report
- **6:30-7:30 PM:** Integration, quality check, submission

DATASET

Source: UCI Machine Learning Repository, "Diabetes 130-US Hospitals for Years 1999-2008" (Strack et al., 2014), derived from the Cerner Health Facts clinical data warehouse

What: Inpatient encounters of patients with a diabetes diagnosis at 130 US hospitals; each row is one hospital stay of 1-14 days with lab tests and medications administered

Size: 101,766 encounters, 71,518 unique patients, 50 columns

Files (provided in the project folder, no download or account needed):

- diabetic_data.csv - the encounters (missing values coded as "?"; "None" in the two lab-result columns means the test was not performed)

- ids_mapping.csv - descriptions for the admission type, discharge disposition and admission source codes

Target: readmitted, with values <30 (readmitted within 30 days), >30 (readmitted after 30 days) and NO. You will model readmission within 30 days.

Backup URL (public): https://archive.ics.uci.edu/dataset/296

The data dictionary explains every column, the ID codes, and the known data quality issues. Read it before writing code.

PROJECT DELIVERABLES

1. Interactive Dashboard (40%, 120 points)

Format: interactive HTML (Plotly), Jupyter Notebook, R Markdown or flexdashboard, Dash, Streamlit, or Shiny

Required Components:

- Executive summary with key metrics (cohort size, readmission rate, model AUC, readmissions captured at your chosen capacity)
- 10-12 interconnected visualizations covering all 10 analyses
- Clinical insights highlighted next to the charts that support them
- Professional design, consistent colors, clear navigation

2. Slide Deck (30%, 90 points)

Format: PowerPoint (.pptx), 10-15 slides, written as if presenting to the hospital's Chief Medical Information Officer and Chief Quality Officer. No live presentation is required, but the slides must stand on their own.

Required Content:

- Title slide with team name and members
- The clinical problem: why 30-day readmission matters clinically and financially (1-2 slides)
- Data and cohort: what you included, excluded and why (1 slide)
- Methods: features, models, evaluation protocol (1 slide)
- Key findings with visualizations (5-7 slides)
- Fairness audit and deployment plan: who gets flagged, what happens next, how you will monitor it (2-3 slides)
- Limitations, risks and recommendation: deploy, pilot, or do not deploy (1 slide)

3. Technical Report (20%, 60 points)

Format: 4-6 pages (PDF or Word), plus appendices

Required Sections:

- Executive summary (1 paragraph)
- Data, cohort definition and preparation (with the cohort funnel)
- Analytical approach: features, models, validation, metrics and why you chose them
- Results: exploratory findings, model comparison, explainability

- Fairness audit: findings and what you would do about them
- Deployment plan: workflow integration, capacity, intervention, monitoring, governance, regulatory and privacy considerations
- Limitations and future work
- Appendix A: Model card (template below)
- Appendix B: AI Tool Use Disclosure with screenshots
- Appendix C (optional): additional charts, code snippets

4. Code Repository (10%, 30 points)

- Clean, commented code organized by analysis
- README.md with setup instructions, how to run, and a file list
- requirements.txt (Python) or package list (R)
- Runs without errors; total run time under 15 minutes
- Output CSVs: cohort_definition, feature_dictionary, model_results, feature_importance, fairness_audit, capacity_analysis, project_summary

REQUIRED ANALYSES

Your group must complete all 10 of the following analyses. The starter code has Analyses 1-6 partly built (with decisions left to you) and Analyses 7-10 as structured TODOs.

Analysis 1: Data Quality and Cohort Definition

- Decide how to handle repeated patients (rows are not independent)
- Exclude encounters where readmission is impossible (died, hospice) and justify any other exclusions
- Handle missing values; note that "missing" is often informative
- Define the target (readmitted within 30 days) and report the positive rate
- Deliverables: cohort funnel chart, cohort_definition.csv

Analysis 2: Demographics and Readmission Rates

- Readmission rate by age group, race and gender with group sizes
- At least one additional demographic view
- Deliverable: analysis2_demographics.html

Analysis 3: Utilization and Clinical Factors

- Prior inpatient, emergency and outpatient visits; length of stay; number of diagnoses, medications and procedures
- HbA1c testing and result vs readmission (the original paper's question)
- Deliverable: analysis3_utilization.html

Analysis 4: Diagnoses and Medications

- Group ICD-9 primary diagnoses into clinical categories (function provided)
- Readmission by diagnosis group, insulin order, medication changes and number of diabetes medications

- Deliverable: analysis4_diagnoses_medications.html

Analysis 5: Feature Engineering

- Numeric features, binary flags, grouped and one-hot encoded categoricals (roughly 55-70 features)
- No identifiers, no raw high-cardinality codes, nothing that leaks the outcome
- Deliverables: feature_dictionary.csv, cohort_features.csv, analysis5_feature_correlations.html

Analysis 6: Baseline Model

- Stratified train/test split; class-weighted logistic regression
- ROC AUC, PR AUC, Brier score, sensitivity and precision; ROC and precision-recall curves
- Interpret the coefficients
- Deliverable: analysis6_baseline_model.html

Analysis 7: Advanced Models and Comparison

- At least two additional models (random forest, gradient boosting), 5-fold cross-validation on the training set
- ROC and calibration curves; select a final model and justify the choice
- Deliverables: model_results.csv, analysis7_model_comparison.html

Analysis 8: Explainability

- Permutation importance on the test set; partial dependence for at least one key feature
- Clinical sense check of the top drivers; agreement between models
- Deliverables: feature_importance.csv, analysis8_explainability.html

Analysis 9: Fairness and Bias Audit

- At a defined operating point, compare AUC, flag rate, sensitivity, false positive rate and PPV across race, gender and age groups
- Identify groups that are under- or over-flagged relative to their risk and say what you would do about it
- Deliverables: fairness_audit.csv, analysis9_fairness_audit.html

Analysis 10: Deployment and Clinical Integration

- Capacity analysis: readmissions captured, PPV and number needed to flag at 5-50% outreach capacity; cumulative gains chart; risk tiers
- Model card (Appendix A of the report)
- Workflow: where the score appears, who acts, what the intervention is, how success is measured, how the model is monitored, who is accountable, and what regulatory and privacy rules apply
- Deliverables: capacity_analysis.csv, analysis10_deployment.html, model_card.md

TECHNICAL REQUIREMENTS

Data Preparation Checklist

- ☐ Load both files; treat "?" and "None" correctly
- ☐ Resolve repeated patients before splitting
- ☐ Exclude encounters where readmission is impossible
- ☐ Group high-cardinality codes (diagnoses, discharge, admission source, specialty)
- ☐ Document every exclusion and grouping decision in the report

Modeling Standards

- Set random seeds; same split for every model
- Cross-validate on training data only; report test-set metrics once
- Use metrics suited to a 9% positive rate: ROC AUC, PR AUC, calibration, sensitivity at a stated operating point. Accuracy alone is not acceptable
- Expect ROC AUC in the 0.62-0.68 range. Published models on this dataset land there. A much higher number almost always means leakage; find it

Minimum Visualization Requirements

- At least 10 distinct visualizations across the 10 analyses
- At least 1 cohort funnel, 2 model-evaluation charts (ROC, PR or calibration), 1 explainability chart, 1 fairness chart, 1 deployment chart
- Consistent color scheme; clear titles, labels and legends

CLINICAL AND BUSINESS CONTEXT

The Scenario

A multi-hospital health system loses several million dollars a year to Medicare penalties under the Hospital Readmissions Reduction Program and wants to target its limited transitional-care nurses at the patients most likely to bounce back. The system can realistically follow up about 15% of discharges. Leadership wants to know: can a model pick the right 15%, is it fair, and what would it take to run it safely?

Key Questions to Answer

1. Who is readmitted within 30 days, and what distinguishes them?
2. How well can we predict it, and which model should we trust?
3. What drives the predictions, and does that make clinical sense?
4. Does the model treat patient groups equitably at the operating point we would actually use?
5. If we flag the top 15%, how many readmissions do we catch, and at what cost in false alarms?
6. Should the system deploy this, pilot it, or not? Under what conditions?

MODEL CARD TEMPLATE (Appendix A)

Section	What to include
Model details	Model type, version, features, training data, date, authors

Section	What to include
Intended use	Primary use case, intended users, decisions it informs, out-of-scope uses
Training data	Source, years, cohort rules, exclusions, size, positive rate
Evaluation	Test set description; ROC AUC, PR AUC, calibration; operating point and its sensitivity, PPV, flag rate
Fairness	Subgroup performance table; known disparities; mitigations considered
Limitations	Data age, missing variables (social determinants, adherence), generalizability, label definition
Monitoring	Metrics tracked, cadence, drift thresholds, retraining triggers, who reviews
Governance	Clinical owner, technical owner, escalation path, how clinicians can override

AI TOOL USE POLICY

Generative AI tools (ChatGPT, Claude, Gemini, Copilot and similar) are **permitted** for three purposes only: (1) debugging error messages, (2) ideating and asking conceptual questions, and (3) cleaning up the grammar and clarity of text and slides you wrote. They are **not permitted** for writing code. The point of this project is to learn the hard coding mechanics and building blocks you will need to build stronger AI implementations.

The technical report must include an **AI Tool Use Disclosure** appendix stating which tools the group used, what for, with screenshots. If none, write "No AI tools were used." Undisclosed AI use, or AI-generated code, is an academic integrity violation for the whole group.

COMMON CHALLENGES AND SOLUTIONS

Challenge 1: The model is bad (AUC 0.65). Solution: it is not bad, it is realistic. Readmission depends heavily on things not in the record. Your job is to show what a modest model can still do operationally (capacity analysis) and to be honest about it.

Challenge 2: The model is suspiciously good (AUC 0.90+). Solution: you leaked. Common culprits: the same patient in train and test, a feature derived from readmitted, or discharge codes that encode death.

Challenge 3: Random forest takes forever. Solution: min_samples_leaf (min.node.size) of 20+, 200-300 trees, use all cores. Cross-validate the slow models with 3 folds if needed and say so.

Challenge 4: Fairness results look noisy for small groups. Solution: require at least 100 test patients per group, report n, and say what the noise means for the conclusion.

Challenge 5: Group coordination. Solution: agree on the cohort and feature matrix FIRST (Analyses 1 and 5) and save cohort_features.csv; every later analysis reads from that file so people can work in parallel. Use a shared drive or repository, check in hourly, and freeze the data by 1:30 PM.

Challenge 6: Running out of time. Solution: finish all 10 analyses at a basic level before polishing any of them. A complete, honest project beats a beautiful, incomplete one.

GRADING RUBRIC (300 POINTS)

Deliverable	Points	Criteria for full credit
Dashboard: analytical completeness	60	All 10 analyses represented correctly; cohort rules, modeling protocol and fairness audit are sound; numbers reconcile across charts and tables
Dashboard: insight and design	40	Executive summary metrics; insights stated next to the evidence; consistent colors; professional layout; interactive elements work
Dashboard: files and reproducibility	20	All required CSV outputs present and consistent with the charts
Slide deck: narrative and recommendation	50	Clear clinical problem; findings quantified; fairness and deployment addressed honestly; an explicit deploy / pilot / do-not-deploy recommendation with conditions
Slide deck: executive quality	40	Stands alone without a presenter; one message per slide; charts readable; 10-15 slides
Technical report: methods and results	30	Cohort funnel, feature choices, validation protocol and results reported precisely; limitations candid
Technical report: fairness, deployment, model card	25	Subgroup findings interpreted; workflow, capacity, monitoring and governance specified; complete model card
Technical report: AI disclosure and format	5	Disclosure appendix with screenshots; 4-6 pages; citations
Code: quality and organization	20	Runs cleanly; commented; organized by analysis; seeds set; README and package list
Code: run time and outputs	10	Completes in under 15 minutes; writes every required output file

TIPS FOR SUCCESS

- Freeze the cohort and feature matrix early; everything downstream depends on it
- Think "so what" for every chart: what would the CMIO do differently after seeing it?
- Quantify. "Prior admissions matter" is weak; "patients with 2+ prior admissions are readmitted at 18.5% vs 8.1%" is strong
- Treat the fairness audit as a finding, not a checkbox. Explain the mechanism behind any disparity you see
- Write the recommendation slide first, then build the evidence for it
- Save work constantly; test the final deliverables before zipping

DELIVERABLES CHECKLIST

Submit ONE zip file per group containing:

Code:

- ☐ Main analysis script(s) (.py / .R / .ipynb / .Rmd)
- ☐ README.md with setup and run instructions

- ☐ requirements.txt or R package list

Dashboard and outputs:

- ☐ Dashboard file (HTML, notebook, Rmd, or app folder)
- ☐ Individual analysis HTML files (analysis1 through analysis10)
- ☐ CSV outputs: cohort_definition, feature_dictionary, model_results, feature_importance, fairness_audit, capacity_analysis, project_summary
- ☐ model_card.md

Documents:

- ☐ Slide deck (.pptx, 10-15 slides)
- ☐ Technical report (.pdf or .docx, 4-6 pages plus appendices)

File Naming: TeamName_GroupProject.zip

Due: Saturday, 31 OCT 2026, 11:59 PM (residency local time) via Blackboard. One submission per group; list all members on the title slide and in the README.

RESOURCES

Dataset and paper: Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. BioMed Research International, 2014, 781670. https://doi.org/10.1155/2014/781670

Dataset citation (APA): Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999-2008 [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J

Policy context: CMS Hospital Readmissions Reduction Program https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/hospital-readmissions-reduction-program-hrrp

Responsible AI: Model cards (Mitchell et al., 2019) https://arxiv.org/abs/1810.03993 ; Fairlearn diabetes example https://fairlearn.org/main/user_guide/datasets/diabetes_hospital_data.html ; FDA Clinical Decision Support guidance https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software

Python: scikit-learn model evaluation https://scikit-learn.org/stable/modules/model_evaluation.html ; permutation importance https://scikit-learn.org/stable/modules/permutation_importance.html

R: caret https://topepo.github.io/caret/ ; ranger https://cran.r-project.org/package=ranger ; gbm https://cran.r-project.org/package=gbm

Good luck. Build something a hospital could actually use, and be honest about what it cannot do.